

# Sign-Up on Deaf Technologies: Reframing Access, Interaction, and Design

Robin Angelini  
Crip Collective || HCI Group  
TU Wien  
Vienna, Austria  
robin.angelini@tuwien.ac.at

Maartje De Meulder  
HU - University of Applied Sciences  
Utrecht, Netherlands  
maartje.demeulder@hu.nl

Shuxu Huffman  
Computer Science  
Colorado State University  
Fort Collins, Colorado, USA  
Gallaudet University  
Washington, District of Columbia  
USA  
shuxu.huffman@colostate.edu

Raja Kushalnagar  
Gallaudet University  
Washington, District of Columbia  
USA  
raja.kushalnagar@gallaudet.edu

Abraham Glasser  
Technology Access Program  
Gallaudet University  
Washington, District of Columbia  
USA  
abraham.glasser@gallaudet.edu

Oliver Suchanek  
Metalab  
Vienna, Austria  
oliver-jayden.suchanek@student.uni-  
ak.ac.at

## Abstract

Research on technologies for deaf communities has grown in recent years, yet it remains fragmented across disciplines and is largely dominated by hearing, non-signing researchers. Deaf scholars and users remain significantly underrepresented, raising critical questions about whose perspectives guide research and development. This deaf-led Sign-Up seeks to address these issues by bringing together researchers from diverse fields to explore Deaf Tech as an emerging research field. Through interactive activities, including the World Café approach, participants will reflect on current challenges, collaboratively expand the understanding of deaf technologies, and identify directions and strategies for future works. The session invites both deaf and hearing researchers to come together, share experiences and perspectives, and build community around deaf technologies. Together, we will explore and strengthen practices and guidelines for research, design, and innovation in ways that are deaf-centered and critically informed.

## CCS Concepts

• **Human-centered computing** → **HCI theory, concepts and models.**

## Keywords

deaf, technologies

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## 1 Introduction

Research on deaf technologies has been gaining increasing attention in recent years. For the purposes of this Meet-Up Sign-Up proposal, we define *deaf technologies* as any technologies related to or impacting deaf people. However, research on technologies for deaf communities has predominantly been shaped by hearing, non-signing researchers, embedding ableist and audist<sup>1</sup> ideologies in the technologies themselves, and in the discourses surrounding them. Furthermore, as there is currently no established understanding of Deaf Tech as a distinct research field, work is fragmented across multiple disciplines such as Human-Computer Interaction, accessibility computing, and (sign) language technologies, resulting in scattered knowledge and practices.

Despite this growing interest, both deaf researchers, users, and participants remain significantly underrepresented in the research, design, and development of deaf technologies. This underrepresentation raises critical questions about whose perspectives are shaping research, development, and practices.

In response, research by deaf scholars has begun to address these issues, stressing the importance of centering deaf perspectives and mutual exchange between deaf and hearing researchers. This growing body of work includes a) centering deaf perspectives [6, 10], b) DIY projects [7, 9] c) reflecting critically on accessibility and languaging practices [5, 8], d) analyses of systematic bias in AI [4], and e) reframing deaf technologies to center deaf epistemologies [1].

<sup>1</sup>Audism refers to discrimination against DHH people based on the belief that hearing people and hearing culture are superior [2]

## 2 Objectives

The primary goal of the Sign-Up, led by deaf scholars, is to bring together diverse attendees from different fields and backgrounds who are working on deaf-related projects, and to stimulate dialogue among them. Its key focus is to explore Deaf Tech as an emerging research field and theoretical framework, which involves thinking about technologies relating to and impacting deaf communities.

In concrete, we aim for the session to provide an opportunity to:

- Collaboratively expand and transform our understanding of what constitutes deaf technologies.
- Identify strategies to advance deaf technologies through critically informed, deaf-centered approaches.
- Share and reflect on current issues in researching and designing deaf technologies.
- Network and build new collaborations between researchers who may not otherwise have connected.

By doing so, we are dedicated to creating an open space for discussion and exchange. The meet-up seeks to strengthen deaf-led perspectives and understanding of access as a collective responsibility when working on deaf technologies, ultimately informing the research, design, and development in ways that center the needs and realities of deaf people.

## 3 Activities

We aim to make the Sign-Up interactive to foster exchange and collaboration among attendees. The 90-minute session is structured as follows:

**Icebreaker & Introductions (15 min)** Organizers and participants will engage in a short, playful introduction activity to share their interests, backgrounds, and current involvement with deaf technologies. This aims to reduce initial barriers to interaction and helps us get an initial sense of participants' priorities.

**Agenda Framing (10 min)** We will outline the session's goals by briefly introducing "Deaf Technologies" as an emerging field and theoretical framework that participants can actively shape and guide through discussion. Together with the group, we will then identify and prioritise three to four key themes that will structure the World Café (e.g., deaf-led research, bias in research and design, collaboration between deaf and hearing researchers, framing of accessibility, etc.). This structure responds to the limited time available and ensures sufficient depth of discussion.

**World Café (50 min)** The primary method will be the World Café [3], enabling interactive, round-table discussions with pre-defined issues. For each of the selected themes, participants will rotate among tables, each featuring a large poster with two or three guiding questions corresponding to mentioned themes (see supplementary materials for details). Participants will rotate between the themed tables in two or three rounds, adding comments, questions, and ideas directly on the posters. Each table will have a host who stays at that table throughout to briefly summarise previous rounds and ensure continuity. This approach allows documentation of the discussion and cross-pollination of ideas as participants rotate between tables.

**Wrap-Up and Synthesis (15 min)** We will conclude with a plenary discussion in which table hosts briefly report back (2-3 minutes each) on key insights, tensions, and open questions resulting

from the theme. Organisers will then facilitate a short synthesis by clustering related ideas (for example, on a whiteboard or shared slide) and asking participants to identify: (1) shared priorities, (2) promising areas for collaboration, and (3) concrete next steps.

## 4 Organizers

To address the underrepresentation of deaf scholars and to prioritize deaf-led approaches, all organizers are deaf or hard of hearing and work on deaf technologies, from different disciplinary perspectives. Their expertise and lived experience inform the session.

**Robin Angelini** (TU Wien) is a Deaf PreDoc researcher on the ACCESSTECH project. His research interests lie at the intersection of deaf technology, critical access and emerging technologies. In his work, he challenges the predominant ideologies in the design and development of technologies for deaf people, shifting toward alternative socio-technical narratives grounded in deaf epistemologies. He's advocating for deaf-centered design approaches that prioritize deaf lived experiences, desires and needs.

**Maartje de Meulder** (HU University of Applied Sciences Utrecht) is a senior researcher working on the intersection of Deaf Studies, Disability Studies, and language technologies, with a focus on sign language AI. She collaborates with deaf communities and sector organisations to develop critical, community-informed perspectives on the design, evaluation, and governance of sign language technologies, and advocates for deaf-led approaches and responsible deployment. <https://maartjedemeulder.be>

**Shuxu Huffman** (Gallaudet University) is a Deaf PhD student and researcher specializing in XR accessibility, with a background in software engineering. Their work investigates how immersive technologies can address long-standing barriers in deaf education, particularly the challenge of visual attention split in classrooms.

**Raja Kushalnagar** (Gallaudet University) is a senior researcher, with over twenty years of experience in the accessible technology field. He collaborates with consumers, industry, and policymakers on AI and accessibility issues, with focus on a deaf/hard of hearing perspectives and evidence-based research. He also investigates information and communication differences between hearing and deaf people. He advocates for laws and regulations to incorporate accessible computing advances in society.

**Abraham Glasser** (Gallaudet University) is an Assistant Professor in the Accessible Human-Centered Computing and Policy (AHCP) program at Gallaudet University, where he is also co-director of the Rehabilitation Engineering Research Center on Technology for the Deaf and Hard of Hearing (DHH RERC), and a faculty researcher in the Artificial Intelligence, Accessibility and Sign Language Center (AIASL). He is also a member of the the Coalition for Sign Language Equity in Technology (CoSET) and has contributed to published resources supporting standards work, e.g. for AI-based interpreting. Overall, he and his students investigate accessible computing for people who are Deaf or Hard of Hearing, including AI, sign language, speech, sign language, and immersive technologies.

**Oliver Suchanek** (they/them) currently majors in Transmedia & Design at the Angewandte (University of Applied Arts, Vienna). They have strong ties to the hacker and maker communities and have curated and exhibited art works relating to deaf identity and

more. Their interest lies in understanding how different technologies allow people to identify potentials for themselves spanning across traditional fields such as Human-Computer Interaction, Interactive Media, and Emergent Technology. As a Deaf curator, they have innovated on access as a form of artistic expression, currently working on an exhibition decidedly addressing and working with the DeafBlind Community.

## 5 Communities

For the Sign-Up, we invite all deaf and hearing researchers, both signing and non-signing, who are working on technologies related to deaf communities or sign language technologies. Participants may come from diverse disciplines, including Human-Computer Interaction, Accessibility Computing, and (Sign) Language Technologies (Sign Language Machine Recognition and Translation), but not limited to those. Hence, our goal is to bring together and establish a community focused on deaf technologies, stressing the importance of acknowledging power dynamics, mutually exchanging perspectives, and adopting community-led approaches. We aim to use this meetup as a starting point for future workshops, grant proposals, position and policy papers that develop actionable approaches to address these challenges.

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